AMAZON BASE BUILDING COMPLIANCE STANDARD

Task Title	Transformer oil testing
APM Object	CT.ELEC.TROILTEST
APM Description	Transformers containing oil
General tasks always required	Follow safety procedures - e.g. 1MRA, LOTO, PPE, etc. Ensure that all WOs are carried out by qualified, certified staff with appropriate familiarisation and supervision as required Add notes for each observation and create a FWO if necessary to eliminate defects Use appropriate tools, chemicals and industry standard practices (RME policy permitting) Ensure relevant logs, readings and certification is uploaded to APM prior to WO closure. CT evidence must be additionally uploaded to the DMS (WorkDocs in DE) To suggest improvements please raise a SIM or contact document authors Leonardo Vitale (LEOVITAL) Rob Gisby (EWARTGXR) or APM, Benito Sanchez Sanchez de Puerta (BPPUERTA) SIM - Link to SIM template

Last Update: (Version)	ago-23	**Last Update Change Detail:**	Transfer to new template. Rationalisation of EU RME Minimum standards. Alignment with dependant PM. Released as Draft.	**Activities may be covered by No-Go restrictions:**	Not Specified	**Special Certification Required:**	**Delivered By:**
Evidenced part or full by PM:	Not Specified	**Related BBM PM:**	ELEC.HV.SWITCHGEAR	**Related CT Areas:**	Not Specified		

Scope

Any site with operating transformers where oils are used as coolant and for its electrical insulating properties. Equipment owned and operated by third parties (e.g. power company) are not included in prescribed PPM activities.

Using this document

This document is intended as a guide and may not be exhaustive. Whilst this document is based extensively on professional guidance and subject to periodic review, individual sites must be diligent to ensure they are aware of, and comply to, local and changing regulations.
The format of these standards is intended to improve navigation between maintenance and compliance standards and to allow frequent, minor updates to a single living resource. Please avoid offline or printed versions and check version dates and change log.
In the event of becoming aware of errors or omissions in this document, sites must provide detail, including references to relevant source information to their local planner for escalation - SIM Link above.
This document is considered released as draft.
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Maintenance and Compliance Relationship

Where relevant, associated Base Building PM's are listed on the header . In many cases maintenance contributes to, or fulfils compliance requirement. This may not be exhaustive and is intended to assist users to identify relevant maintenance.
If a BBM Maintenance activity fully meets the compliance requirements below, periodic upload of PM evidence will be required to APM and DMS / Workdays depending on region.

Key to Abbreviations

The key to the below document is as follows:
CT - Amazon compliance / statutory task - not otherwise covered by a BBM Standard PM. Evidence to be provided into APM and DMS upon task closure.
CT* - This is a conditional task based or based on findings from previous inspections. If a frequency is not used, a further appropriate frequency needs to be in either the region, or 'ALL'. Local teams must ensure appropriate frequency.
(CT) - Compliance task which is likely met or exceeded by a CT task in [ALL] and therefore achieved by proxy BUT local site must satisfy themselves that requirements are fully met.
PM - Requirement can demonstrated through appropriate maintenance - Refer to relevant PM Standard
(PM) - Maintenance task which is met or exceeded by a PM task in [ALL] and therefore achieved by proxy BUT local site must satisfy themselves that requirements are fully met including regional / manufacturer requirements.
EVID - Periodic upload of evidence to DMS, for non-CT task lines. For example, evidence of PM completion, certification, licence etc.
EVID* - Conditional requirement to upload evidence at the interval provided
N A R - No Additional Requirement

Required Tasks

Region	Task Index	Task Frequency	Task Requirements	Qualification of Inspector	Records Required	PM	Weekly	Monthly	3 Monthly	6 Monthly	12 Monthly	24 Monthly	Additional
All	1	In accordance with manufacturer's instructions and recommendations. Based on risk assessment (operational and environmental conditions). Generally performed at least every 12 months.	Regular transformer oil testing undertaken to identify potential issues at an early stage, including visual inspection, acidity, dielectric constant, density, water content, etc.	Competent person	Analysis results						CT		
All	2	In accordance with manufacturer's instructions and recommendations. Based on risk assessment (operational and environmental conditions). Generally performed at least every 12 months.	Where transformers pre-date 1986, Poly Chlorinated Biphenyls (PCB) analysis should have already been completed.	Competent person	Analysis results						CT		
All	3	During maintenance of the equipment, and following an oil change, disposal and refurbishment.	Old, unlabelled transformers and those manufactured before 1986 could potentially contain PCBs. Transformers with PCBs of less than 0.05% by weight are still allowed to be used, however maintenance frequency should be reviewed.	Competent person	PCB-containing equipment register. Analysis results								On demand EVID

Related Regional Legislation

Region	Relevant legislation	Vendor Certification	Statutory requirement
ALL	No region specific legislation specified	N/A	N/A

CZ	- ČSN 35 1090 - ČSN 33 1500 Decree No. 50/1978 Coll.	Competent person - qualified electrician	Inspections according to the preventive maintenance specified by the employer.
DE	DIN EN 60422 VDE 0370-2 - Isolieröle auf Mineralölbasis in elektrischen Betriebsmitteln - Leitlinie zur Überwachung und Wartung	Competent person	Oil testing frequency is decided by the operator and depends on the type of equipment, its function, voltage, power, stress, the results of previous examinations, and the effects that a failure of the operating equipment has on the company.
ES	RD 228/2006, of February 24, which modifies RD1378/1999 and establishes regulation for the elimination and management of polychlorinated biphenyls, polychlorinated terphenyls and all the devices where are contained. RD 337/2014, of May 9, which approves the high voltage electric systems and its 23 Complementary Technical Instructions. UNE-EN 60422 Isolating oils in electrical equipment	Accredited company (OCA) by ENAC.	High Voltage inspections and testing to be conducted at least every 3 years.
FR	Equipment manufactured after 18 June 1994 is considered to be free of PCB contamination https://aida.ineris.fr/consultation_document/24152 For equipment built before 1987, without free PCB label, an oil analysis is required to determine PCB concentration https://www.legifrance.gouv.fr/codes/article_lc/LEGIARTI000027298297 https://www.legifrance.gouv.fr/loda/id/JORFTEXT000028478731/	Competent person, trained to undertake sampling and analysis.	No Frequency defined by French regulation.
IT	CEI EN 60422 (Mineral insulating oils in electrical equipment- Supervision and maintenance guidance)	Competent person	Dielectric strength analysis, gas chromatography of the oil, if the PCB level is superior than 50 p.p.m report to local authorities (ASL). Frequency depends on equipment and operating conditions, best practice is every 12 months.
PL	The Act of April 27, 2001, Environmental Protection Law. (Journal of Laws of 2001, No. 62, item 627) Regulation of the Minister of Economy of 24 June 2002. On the requirements for the use and movement of substances posing a particular threat to the environment and the use and cleaning of installations or devices in which they were or are used substances posing particular environmental threat. (Journal of Laws of 2002, No. 96, item 860) Regulation of the Minister of Economy of September 26, 2002 on the determination of devices in which substances posing a particular threat to the environment could be used (Journal of Laws of 2002, No. 173, item 1416)	Competent person, trained to undertake sampling and analysis.	From June 30, 2010, it is not allowed to use PCBs in devices or installations in use.
ROI	BS EN 60422:2013 - Mineral insulating oils in electrical equipment. Supervision and maintenance guidance	Competent person	Frequency depends on equipment and operating conditions.
SK	Act No. 124/2006 Coll. on safety and health at work Ordinance No. 508/2009 Coll. on details for ensuring safety and health protection at work with pressure, lifting, electrical and gas technical equipment STN EN 60076-2: Power transformers. Part 2: Temperature rise	Certified and qualified person	Every two years for objects containing flammable materials.
UK	BS EN 60422:2013 - Mineral insulating oils in electrical equipment. Supervision and maintenance guidance	Competent person	Frequency depends on equipment and operating conditions, but analysis is generally undertaken at least every 12 months.